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import cv2
import numpy as np
from google.colab.patches import cv2_imshow

# Load the image from the Colab environment
image = cv2.imread('image1.jpg')

if image is not None:
    # Laplacian mask with a POSITIVE center coefficient for sharpening
    # This kernel effectively performs: Original + (Original - Smoothed)
    # Common 3x3 sharpening kernel with positive center:
    kernel = np.array([[0, -1, 0],
                       [-1, 5, -1],
                       [0, -1, 0]])

    # Apply the filter directly to get the sharpened image
    sharpened = cv2.filter2D(image, -1, kernel)

    # Display results
    print("Original Image:")
    cv2_imshow(image)
    print("\nSharpened Image (Positive Center Laplacian Mask):")
    cv2_imshow(sharpened)

    # Save the result
    cv2.imwrite('Sharpened_Positive_Center.jpg', sharpened)
else:
    print("Error: 'image1.jpg' not found. Please ensure the image is uploaded")
```

Original Image:



Sharpened Image (Positive Center Laplacian Mask):

